

Pretraining a foundation model for small-molecule natural products

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Natural products, as metabolites from microorganisms, animals or plants, exhibit diverse biological activities, making them crucial for drug discovery. Nowadays, existing deep-learning methods for natural products research primarily rely on supervised learning approaches designed for specific downstream tasks. However, such one-model-for-a-task paradigm often lacks generalizability and leaves substantial room for performance improvement. Additionally, existing molecular characterization methods are not well-suited for the unique tasks associated with natural products. Here, to address these limitations, we pretrained a foundation model for natural products (NaFM) based on their unique properties. Our approach employs a pretraining strategy specifically tailored to natural products. By incorporating contrastive learning and masked graph learning objectives, we emphasize evolutionary information from molecular scaffolds while capturing side-chain information. NaFM achieves state-of-the-art results in various downstream tasks related to natural product mining and drug discovery. We first compare taxonomy classification with synthetic molecule-focused baselines to demonstrate that current models are inadequate for understanding natural synthesis. Furthermore, by diving into a fine-grained analysis at both the gene and microbial levels, NaFM demonstrates the ability to capture evolutionary information. Eventually, our method is applied to virtual screening, illustrating informative natural product representations that can lead to more effective identification of potential drug candidates.

Natural products (NPs) are metabolites or secondary metabolites produced by bacteria, fungi, animals, plants and other organisms. Their structural diversity is immense, encompassing hundreds of thousands of distinct compounds¹. As metabolites, they are inherently bioactive, making them valuable candidates for direct pharmaceutical development or structural modification into therapeutic agents—particularly for anticancer and anti-infective drugs^{2–4}. Despite their value, discovering natural compounds suitable for drug development remains

time-consuming and costly⁵. Traditional bioactivity-guided discovery involves extracting metabolites using solvents of varying polarity, followed by successive bioactivity-guided fractionation until pure active compounds are isolated⁶. This process often requires identifying both the structural class of the compound and the producing microorganism to avoid unnecessary cultivation. Moreover, phenotypic screening of bioactive compounds demands substantial effort to determine their molecular targets⁷. General small-molecule databases,

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such as ZINC⁸ and Enamine REAL⁹, enable large-scale virtual screening. For instance, Stokes et al.¹⁰ leveraged machine learning models to identify new antibiotics from vast structural libraries. Building on this foundation, the NP community has curated extensive domain-specific resources, including comprehensive compound collections^{11–13}, activity and biological-origin databases^{14,15}, repositories of biosynthetic gene clusters (BGCs)¹⁶ and specialized marine NP datasets^{17–19}. Concurrently, machine learning approaches have been developed to mine these resources^{20–25}. For example, Xu et al.²⁶ applied deep learning to the taxonomic classification of NPs. Commonly used NP classification tools, such as ClassyFire²⁷ and Natural Product Classifier (NP Classifier)²⁸, rely solely on supervised learning. However, the limited availability of labelled data and inherent dataset biases often reduce their accuracy and generalization, particularly for out-of-distribution or less-related downstream tasks. Despite these limitations, they remain standard classification tools across many NP databases.

Pretraining on large-scale unlabelled molecular data has emerged as an effective strategy for representation learning²⁹. Early methods borrowed natural language processing techniques, encoding Simplified Molecular Input Line Entry System (SMILES) strings³⁰ with recurrent neural networks³¹ or Transformers³² for masked language modelling^{33,34}. Subsequent studies showed that graph neural networks (GNNs)^{35,36} and graph-specific pretraining strategies^{37–39} yield superior results, whereas integrating two-dimensional graphs with three-dimensional (3D) structures, semantic information or force-field data further enhances performance^{40–43}.

Despite these advances, few foundation models target NPs, which differ fundamentally from synthetic molecules in structural diversity and complexity⁴⁴. Their unique scaffolds, stereochemistry and functional groups challenge existing 3D or force-field-based approaches. Yet, NPs exhibit a distinctive organizing principle: analogous to the central dogma of molecular biology, intrinsic relationships link their biological origins, scaffold architectures and biochemical properties. Similar compounds are more likely to originate from related biological sources that share overlapping biosynthetic machinery. In particular, NPs with comparable scaffolds or functional motifs often arise from similar genes or BGCs, which encode enzymes or protein substrates that direct specific metabolic pathways, leading to conserved structural frameworks.

For instance, the mevalonate pathway commonly produces terpenoid and steroid scaffolds, and compounds sharing identical scaffolds often exhibit similar biochemical properties⁴⁵. This intrinsic relationship links upstream genetic clusters and biosynthetic pathways to downstream bioactivity, as illustrated in Fig. 1d. Consequently, conventional molecular graph learning methods that distribute learning objectives across entire molecules are insufficient for modelling NPs. A more effective approach is to adopt a hierarchical pretraining paradigm centred on scaffolds while capturing side-chain structures to organize the overall representation space of NPs.

Technically, current pretraining strategies face inherent limitations. In molecular contrastive learning (CL), data augmentation typically involves masking or removing molecular substructures^{37,40}. However, unlike images, chemical structures are highly sensitive to such perturbations—minor structural changes can cause large variations in chemical or biological activity, a phenomenon known as activity cliff^{46–48}. Consequently, molecules with small structural differences cannot be treated as positive pairs⁴⁹. Furthermore, distinct molecules may yield structurally similar or even identical negative samples, violating the assumptions of CL. Moreover, current masked graph modelling approaches face limitations. Masking a single atom or bond allows the model to infer missing information from adjacent nodes and edges, limiting its ability to capture global structure. Ideally, the model should consider broader molecular contexts when predicting masked components. In addition, access to complete topological information enables shortcut learning, which may weaken the intended training objectives. Studies have shown that increasing pretraining difficulty

improves effectiveness⁵⁰. Therefore, refining masking strategies and introducing additional prediction targets can substantially enrich the model's learned representations.

In this study, we pretrained a foundation model for small-molecule NPs (NaFM), which seamlessly combines masked learning and CL, as shown in Fig. 1a. For the masked learning task, we introduce an approach that extends beyond masking categorical features of atoms and bonds. Specifically, we also mask atomic connectivity, effectively concealing the topological structure of subgraphs. To reconstruct these subgraphs, the model must first infer bond locations and subsequently determine their types. Recognizing that the core scaffold reflects biosynthetic logic and that structural diversity arises from enzyme-mediated modifications before or after scaffold assembly, we distinguish biosynthetic features at the scaffold level from those of peripheral substituents. Accordingly, scaffold similarity is used as a weighting function in our contrastive pretraining. This soft-weighting strategy allows the model to distinguish strong from weak negative examples based on scaffold similarity while incorporating informative signals from side chains. To integrate contrastive and masked learning into a unified framework, we design the contrastive set-up such that positive examples are masked molecules generated during the masked learning process, whereas negative examples come from other molecules and their masked variants. This mutual supervision between the two paradigms enhances representation learning and improves overall performance.

By tailoring pretraining to NPs, NaFM supports diverse downstream applications. Fine-tuned on the NP Classifier dataset²⁸, it achieved superior accuracy in NP taxonomy classification. NaFM embeddings effectively captured biosynthetic relationships in NPs occurring in the LOTUS database¹³, clearly separating metabolites by biological origin. Using BGCs from the MIBiG repository¹⁶, the model enabled genomic mining directly from metabolite structures, identifying biosynthetic genes and protein families. In protein–ligand binding affinity prediction, NaFM outperformed existing models. Additionally, in a case study on acetylcholinesterase (AChE), NaFM demonstrated strong virtual screening ability, with prioritized molecules showing higher inhibitory activity than controls.

Results and discussion

Overview of NaFM

NaFM is a pretraining framework tailored for NPs, integrating two tasks: scaffold–subgraph reconstruction and scaffold-aware CL. As illustrated in Fig. 1b, scaffold–subgraph reconstruction employs a subgraph masking strategy that extends beyond atom and bond masking to include topological information. Because NP scaffolds often exhibit complex polycyclic architectures with intricate atom connectivity, masking topology both enriches the learned representation and increases the reconstruction difficulty. To further emphasize scaffold features, the subgraph mask is applied specifically to the scaffold region of each molecular graph.

In CL, transformed objects originating from the same data points are treated as positive samples, whereas those derived from different objects are treated as negative samples. Although this approach is effective in traditional data forms such as images, its application to molecular data, particularly NPs, presents challenges. This complexity arises because substructures from the same molecule can exhibit substantial differences after augmentation, whereas different molecules may share identical substructures. To address potential mislabelling and accommodate the unique characteristics of NPs, we incorporated scaffold similarity as a soft weight, guiding the model to distinguish negative samples based on biosynthetic relationships.

To seamlessly integrate the two training tasks, the masked graphs are used not only for the scaffold–subgraph reconstruction task but also serve as positive examples in CL. In practice, CL generates a $2N \times 2N$ logits matrix, as shown in Fig. 1c, in which the positive example for each complete molecule is its corresponding masked graph, and the

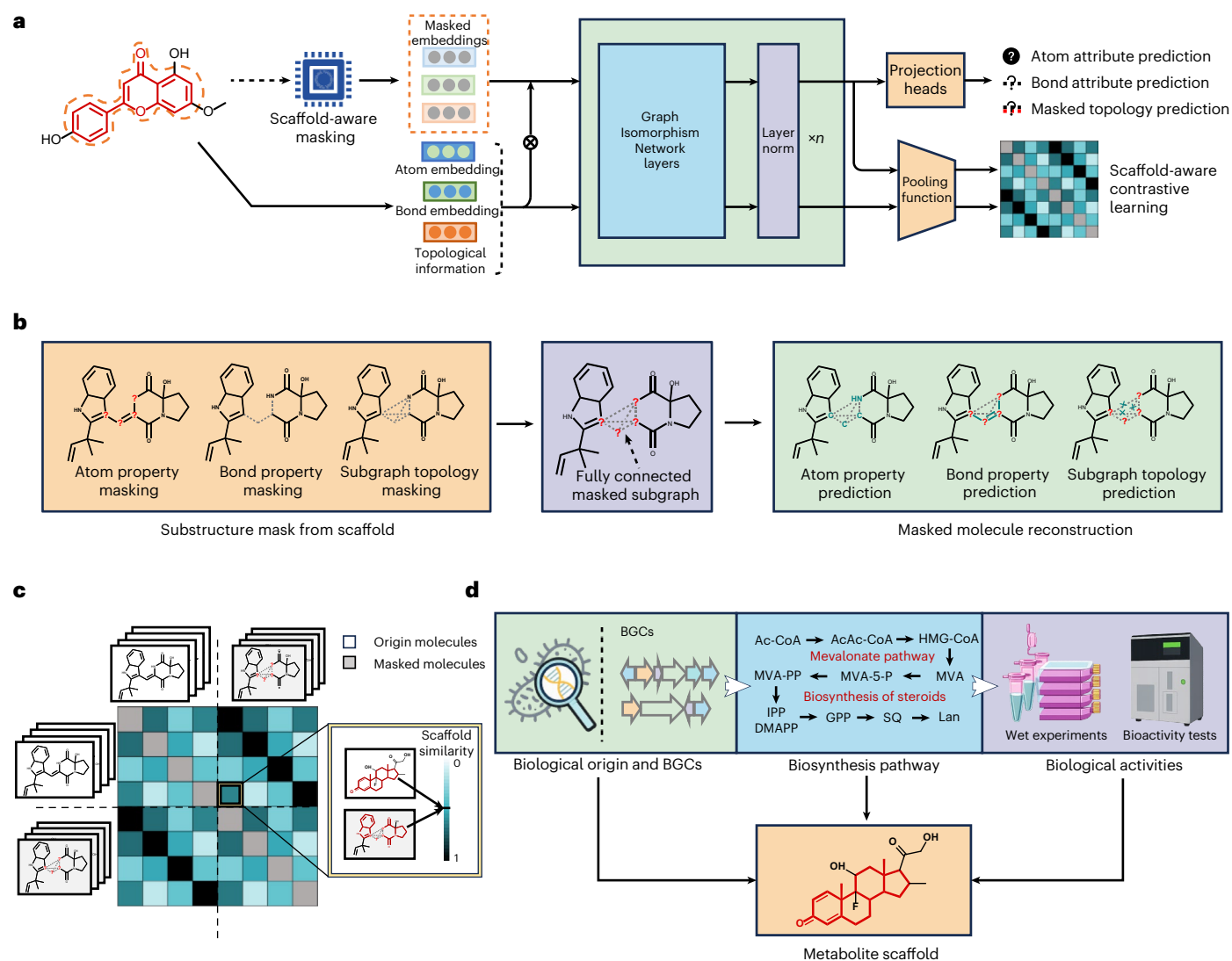


Fig. 1 | Scaffold-aware pretraining framework of NaFM. a, Overview of NaFM pretraining. After the NP molecule undergoes scaffold-aware masking, both the masked information and unmasked information are simultaneously input into a multilayer Graph Isomorphism Network (GIN). The masked information is then processed through a projection head to predict the masked atom attributes, bond attributes and topological information. Meanwhile, the masked information and unmasked information pass through a pooling function and are used for scaffold-aware CL. **b**, Details of scaffold-subgraph reconstruction. First, a subgraph is randomly selected from the scaffold, consisting of multiple atoms and chemical bonds. In the subgraph, both node and edge attributes are masked, and all nodes within the subgraph are fully connected (that is, expanded into a fully connected graph), thereby masking the topological

information. During the reconstruction process, the model needs to predict both node and edge attributes while also distinguishing between real edges and those artificially added virtual edges. **c**, Details of scaffold-aware CL. Different colours represent varying weights defined by scaffold similarity. **d**, The 'central dogma' of NPs. The biological source, BGCs, biosynthetic pathways and bioactivity of NPs are interconnected through the scaffold, which acts as a bridge linking these three key aspects. Ac-CoA, acetyl-coenzyme A; AcAc-CoA, acetoacetyl-coenzyme A; HMG-CoA, 3-hydroxy-3-methylglutaryl-coenzyme A; MVA, mevalonate; MVA-5-P, mevalonate-5-phosphate; MVA-PP, mevalonate diphosphate; IPP, isopentenyl pyrophosphate; DMAPP, dimethylallyl pyrophosphate; GPP, geranyl pyrophosphate; SQ, squalene; Lan, lanosterol.

negative examples are formed by other molecules and their masked graphs. This approach fosters the learning of richer and more accurate molecular representations.

During the pretraining phase, we used ~0.6 million unlabelled data samples from the COCONUT database¹². For downstream tasks, the GNN backbone parameters were retained from pretraining, whereas the prediction head was randomly initialized and fine-tuned together with the backbone on task-specific datasets. Further details of these processes are provided in Methods.

Natural product taxonomy classification

The taxonomy classification of NPs is a classical and important task. A valid NP classification system could be valuable for database mining.

Predicted taxonomy labels can be applied to infer structure-activity relationships, enabling preliminary predictions of their potential biochemical activities.

We mainly evaluated the classification performance on the NP Classifier dataset²⁸, which includes seven biosynthetic pathways of NPs and detailed classifications (SuperClass and Class) based on their structures. After cleaning the dataset by removing labels and corresponding data that were too sparsely represented, we obtained ~77,000 entries, with labels spanning 7 types of biosynthetic pathways, 70 SuperClasses and 563 Classes.

We first fine-tuned NaFM at the Class level. Several pretrained molecular models were replicated and fine-tuned on the same task for comparison, with the non-pretrained Extended-Connectivity

Table 1 | AUPRC results for NP taxonomy classification compared with other baselines

Training samples per Class	4	8	16	24	40	64
N-Gram ⁷⁴	44.72 ± 1.91	56.61 ± 0.66	66.36 ± 1.57	71.11 ± 0.77	73.34 ± 1.14	75.77 ± 0.54
PretrainGNN ³⁸	44.83 ± 0.82	61.85 ± 0.78	75.76 ± 0.62	80.50 ± 0.17	85.31 ± 0.36	87.82 ± 0.13
D-MPNN ⁷⁵	46.63 ± 0.23	60.88 ± 0.48	75.73 ± 0.12	80.96 ± 0.24	86.64 ± 0.44	89.23 ± 0.79
MolCLR ³⁷	45.76 ± 1.98	65.80 ± 1.51	78.14 ± 1.08	83.20 ± 0.74	85.56 ± 0.11	88.22 ± 0.34
Mole-BERT ³⁹	66.32 ± 1.23	73.39 ± 0.66	78.25 ± 0.45	80.83 ± 0.51	83.57 ± 0.59	85.69 ± 0.70
ECFP ⁷⁶	69.17 ± 0.19	78.21 ± 0.79	83.82 ± 0.37	86.28 ± 0.49	88.52 ± 0.48	89.75 ± 0.45
GraphMVP ⁴⁰	64.50 ± 0.74	78.41 ± 0.09	85.71 ± 0.43	87.88 ± 0.26	89.72 ± 0.45	91.07 ± 0.43
NaFM	70.10 ± 0.92	79.89 ± 0.07	87.37 ± 1.51	89.15 ± 0.22	90.77 ± 0.26	91.75 ± 0.47

AUPRC values were computed three times using different random seeds. The results are reported as mean ± s.d. Bold values indicate the best performance in each column.

Fingerprint (ECFP) included as an additional baseline. Datasets were split into training, validation and test sets using stratified sampling to preserve class balance. To assess robustness under varying data scales, we further limited the number of molecules per class in the training set. As shown in Table 1, NaFM consistently outperformed all baselines across different test sizes, with a more pronounced advantage on smaller datasets. GNN-based methods such as molecular contrastive learning of representations (MolCLR), pretraining of graph neural network (PretrainGNN) and directed message-passing neural network (D-MPNN) show strong dependence on fine-tuning data volume, with performance dropping sharply as data become scarce. In contrast, the molecular adaptation of bidirectional encoder representations from transformers (Mole-BERT) benefits from its pretrained tokenizer, which distinguishes atoms of the same type through multiple learned tokens. Overall, NaFM is the only pretrained model that surpasses the rule-based ECFP, which is the previous common choice for NP classification.

Beyond comparisons with general molecular learning frameworks, we benchmarked NaFM against domain-specific tools for NP classification. NP Classifier was selected for its open-source accessibility, scientifically grounded taxonomy and broad adoption in the field^{12,15,19}. We fine-tuned the model on the NP Classifier dataset and evaluated performance at both the Pathway and SuperClass levels. As shown in Fig. 2, NaFM consistently outperformed the NP Classifier across all biosynthetic pathways and finer SuperClass categories. At the pathway level, it improved amino acid and peptide recognition by 24%. At the SuperClass level, NaFM surpassed the NP Classifier in 53 of 71 classes, achieving over 20% gains for Carotenoids (C45), Tetramate alkaloids, Peptide alkaloids and Anthranilic acid alkaloids. The small gap for γ -lactam- β -lactones arises from data scarcity—only eight examples among 78,000 molecules, yielding a single test instance. These results establish NaFM as the state-of-the-art accurate framework to date for NP classification across diverse NP databases.

Biochemical significance of NaFM representation

In this section, we explore the biological information integrated within our pretrained models. Given the strong interrelationship between molecular scaffolds, microbial species and BGCs, the specially designed pretrained model performed exceptionally well in addressing both tasks. We fine-tuned the model using the LOTUS dataset¹³, which contains 130,000 data points with biological source labels. As illustrated in Fig. 3a, NaFM effectively separated NPs from five major biological sources (animals, bacteria, chromista, fungi and plants) within the representation space. Research on chromista is still emerging and has not kept pace with studies on other biological sources. Their unique environments and the difficulties in obtaining samples have resulted in few NPs being identified from them, with only about 1% of data coming from chromista. Nevertheless, despite the scarcity of training data, NaFM was still able to effectively differentiate NPs from chromista. For each biological source, we performed structural clustering and

selected the representative scaffold corresponding to the cluster with the most data. This scaffold reflects the most characteristic structure of NPs from that source. The distinct differences in scaffolds across biological sources further highlight the strong relationship between the source and the scaffold structures. In contrast, applying the same model architecture without pretrained weights failed to distinguish NPs from different sources (full-sized non-pretrained Atlas shown in Supplementary Fig. 13).

In addition, we conducted a comprehensive follow-up analysis to investigate the structural and biochemical underpinnings of the learned embeddings, as detailed in Supplementary Section 2.4. This extended analysis includes confusion-matrix-based error inspection, identification of shared scaffolds across biological sources and case studies of structurally ambiguous or misclassified compounds. These results revealed that most misclassifications stem from genuine biochemical overlap, such as common terpenoid and indole scaffolds shared by plants, fungi and animals. In addition, we generated atlases using different baselines under various training scenarios to further validate the ability of NaFM to classify biological sources (Supplementary Fig. 16). Moreover, to demonstrate the capability of NaFM in distinguishing finer-grained biological classifications, we also produced a heat map combined with a phylogenetic tree organized at the levels of domain, kingdom, phylum and class (Supplementary Fig. 17). These results provide strong evidence that NaFM effectively captures biologically meaningful distinctions.

Genetic studies have shown that bioactive NPs are synthesized by co-localized genes organized into BGCs^{51,52}. Although traditional methods predict metabolites from BGC structures^{24,53,54}, reverse prediction—from metabolite structures to BGCs—enables precise localization of genomic regions responsible for biosynthesis, improving cluster selection and large-scale production. To evaluate this capability, we assembled a dataset of over 2,000 bacterial and fungal BGCs from MIBiG¹⁶ and their associated protein families from Pfam⁵⁵. NaFM and several baseline molecular representations were trained to predict BGC-encoded protein families from metabolite structures. As shown in Fig. 3b, we analysed 128 frequent protein families (Supplementary Table 10). NaFM outperformed MolCLR, ECFP and PretrainGNN in nearly all categories.

Specifically, analysis of the blue blocks in the figure revealed that the model showed markedly better classification performance on certain protein families, such as the Conserved Protein Domain Family (ID: 65 in the figure) and HATPase_c (ID: 115 in the figure), compared with other baselines. The Conserved Protein Domain Family includes protein domains that are widely conserved across different species and are involved in a variety of biological processes, indicating structural and functional similarity⁵⁶, whereas the HATPase_c family is associated with ATP hydrolysis, playing a key role in cellular processes such as signal transduction and DNA repair⁵⁷. Both families are considered conserved protein families. These results suggest that NaFM is capable of making effective predictions by learning the conserved features

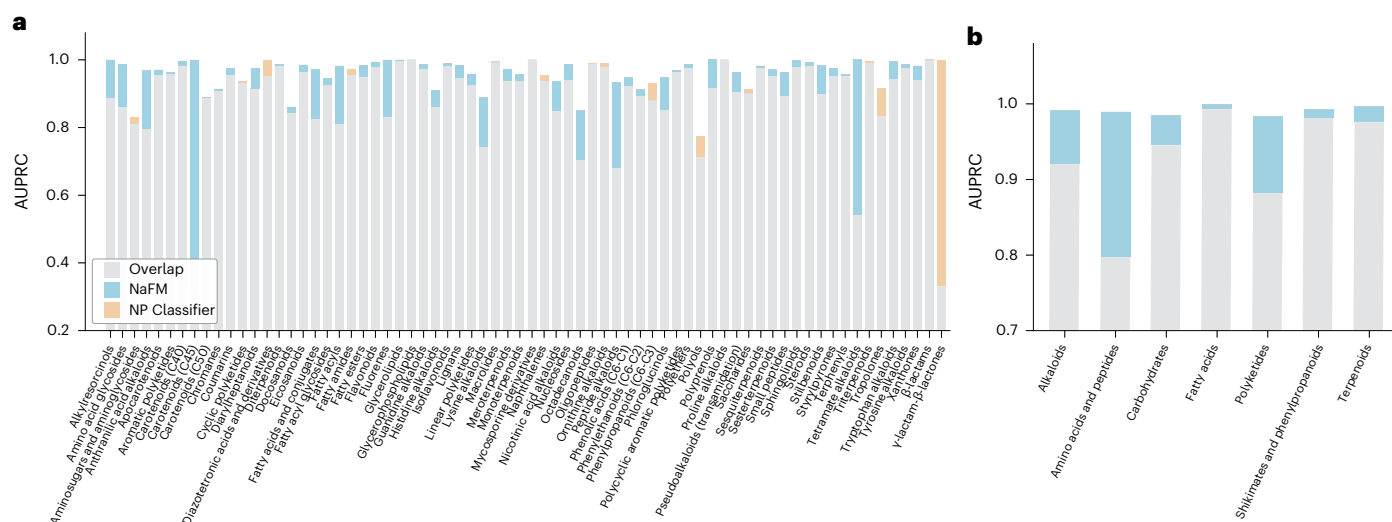


Fig. 2 | Comparison of NaFM and NP Classifier across SuperClass categories and biosynthetic pathways. a, b, Bar plots show model performance in terms of AUPRC for SuperClass categories (a) and biosynthetic pathways (b), including carbohydrates, amino acids and peptides, alkaloids, terpenoids, shikimates and phenylpropanoids, polyketides and fatty acids. Grey segments indicate shared performance (minimum AUPRC), whereas light blue and orange

segments represent the additional AUPRC achieved by NaFM and NP Classifier, respectively. NaFM outperformed NP Classifier across all biosynthetic pathways and most SuperClass categories, with improvements of up to 24% for amino acids and peptides. At the SuperClass level, NaFM achieved higher AUPRC in 53 of 71 classes, with particularly strong gains in carotenoids, alkaloids and peptide-related categories.

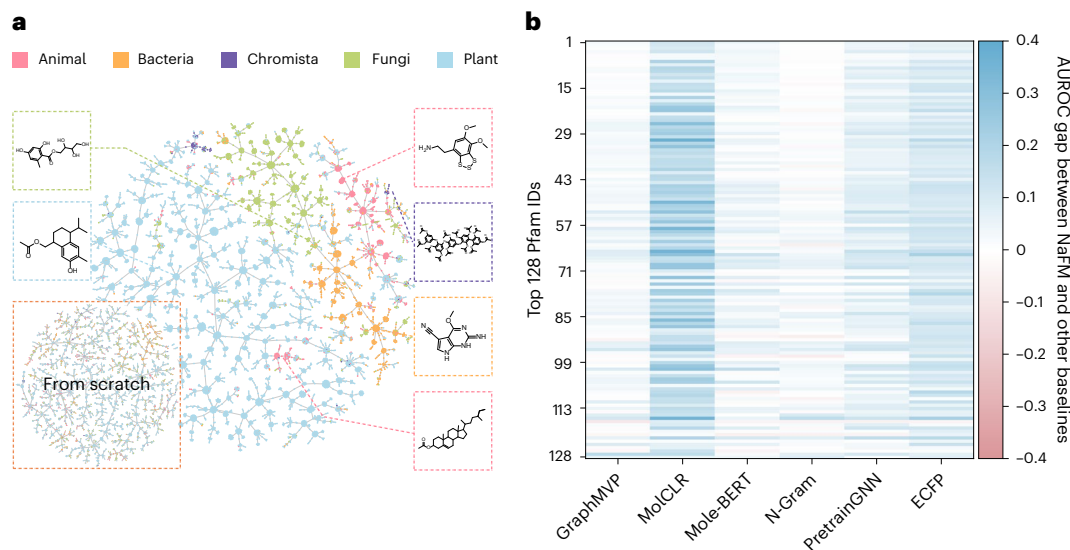


Fig. 3 | Representation learning enables biological source discrimination and reverse BGC functional prediction. a, Natural Products Atlas after representation learning. Compounds from the Natural Products Atlas were embedded using the pretrained and fine-tuned molecular representations and projected into two dimensions. The learned representation organizes chemical space into distinct clusters corresponding to biological origins (animal, bacterial, chromistan, fungal and plant), revealing clear source-specific structural signatures. Representative scaffolds are highlighted. A steroid-like scaffold in the lower right illustrates a failure case, as it occurs across multiple

sources and leads to ambiguous placement. The lower-left panel shows the Atlas visualization trained from scratch without pretraining, demonstrating reduced structural discrimination. **b,** AUROC gap for predicting BGC-encoded protein families. AUROC values are shown for 128 most frequent protein families in bacteria and fungi. Colour indicates the magnitude of the AUROC gap (red to blue; deeper colours represent larger absolute differences). Across most protein families, NaFM achieves higher AUROC than baseline representations, with average improvements of 10 points over ECFP and 20–40 points over MolCLR, particularly for families with conserved functional domains, such as HATPase_c.

within these structures. In addition to analysing the top 128 protein families, we also compared the average classification performance across all protein families. As shown in Supplementary Fig. 15, NaFM substantially outperformed the other baselines in terms of both area under the receiver operating characteristic curve (AUROC) and area under the precision–recall curve (AUPRC), with a smaller variance.

Furthermore, to demonstrate the applicability of the genome-mining capabilities of NaFM in real-world scenarios, we conducted a

comprehensive case study in which NaFM was applied to the reverse inference of biosynthetic features of NPs. The detailed experimental procedures and corresponding results are provided in Supplementary Section 2.5.

In summary, NaFM demonstrates exceptional genetic-level predictive performance, establishing it as a powerful tool for future BGC mining and biosynthetic pathway analysis, particularly for NPs that are difficult to synthesize or annotate.

Table 2 | Comparison of r.m.s. error performance for NP bioactivity regression

Target name	PTP-1B	AChE	COX-2	HIV type-1 reverse transcriptase	Tyrosinase	CYP3A4	MRP4	COX-1
Number of samples	612	341	190	186	186	178	177	140
<i>N</i> -Gram ⁷⁴	0.8493 ± 0.2623	1.3774 ± 0.3995	1.0022 ± 0.3192	1.0606 ± 0.1613	0.7822 ± 0.1408	0.7846 ± 0.2037	0.2747 ± 0.0846	0.9529 ± 0.1643
PretrainGNN ³⁸	0.8652 ± 0.2110	1.5462 ± 0.2265	1.7231 ± 0.3754	1.5210 ± 0.2606	1.6957 ± 0.2915	1.4157 ± 0.2923	1.6869 ± 0.2476	2.6831 ± 0.2056
D-MPNN ⁷⁵	0.8577 ± 0.2165	1.2881 ± 0.1992	1.0272 ± 0.2468	1.1826 ± 0.1683	0.8206 ± 0.1891	0.6946 ± 0.1415	0.2399 ± 0.0603	0.9405 ± 0.1414
MolCLR ³⁷	0.9912 ± 0.2467	1.1507 ± 0.1683	0.9399 ± 0.1705	1.2649 ± 0.1353	0.9165 ± 0.2820	0.7072 ± 0.1362	0.2321 ± 0.0847	0.9574 ± 0.1655
Mole-BERT ³⁹	0.8624 ± 0.0732	1.2938 ± 0.3615	1.0529 ± 0.1536	1.4348 ± 0.1398	1.0440 ± 0.2569	0.9377 ± 0.1873	0.5190 ± 0.0645	1.4542 ± 0.3456
ECFP ⁷⁶	1.1409 ± 0.2281	1.3721 ± 0.0703	1.2718 ± 0.1531	1.5446 ± 0.1133	1.2553 ± 0.0985	1.0439 ± 0.1244	1.4682 ± 0.2422	1.1965 ± 0.1850
GraphMVP ⁴⁰	1.8865 ± 0.1303	2.8140 ± 0.3649	3.0341 ± 0.6418	2.7829 ± 0.1195	3.3965 ± 0.3437	3.0817 ± 0.2720	3.5723 ± 0.4337	3.0083 ± 0.2802
NaFM	0.8243 ± 0.1960	1.1227 ± 0.1604	0.9239 ± 0.1721	1.0802 ± 0.1506	0.6927 ± 0.2828	0.6922 ± 0.1326	0.2265 ± 0.0921	0.9286 ± 0.1342

Bioactivity values are expressed as pIC₅₀. The reported r.m.s. error values represent the mean of three independent runs initialized with different random seeds. The results are reported as mean ± s.d. Bold values indicate the best performance in each column.

Natural product bioactivity prediction and virtual screening

Owing to the difficulty in synthesizing or extracting NPs, there are fewer activity data available for them compared with synthetic molecules⁵⁸. Consequently, the knowledge learned through fine-tuning is limited, posing a challenge for the amount of information the model can acquire directly during pretraining. Although cell line data are more abundant, their complexity—stemming from multiple proteins, organelles and signalling pathways—renders the structure–cytotoxicity relationship intricate. In contrast, protein targets with well-characterized 3D structures and annotated binding sites enable clearer structure–activity relationships through defined molecular interactions (for example, hydrogen bonds, hydrophobic contacts and ionic bonds). Therefore, we focused on such structurally resolved protein targets for subsequent activity prediction and screening experiments.

To validate the activity prediction capability of NaFM, we selected human protein targets and HIV type-1 reverse transcriptase activity data from the NPASS NP activity database, with activity data greater than 100 for each target. We then fine-tuned NaFM and various molecular representation baselines on this refined dataset and recorded the r.m.s. error for each method across different protein target activity prediction tasks. The results are shown in Table 2. As demonstrated, NaFM surpassed all baselines, with the exception of *N*-Gram on HIV type-1 reverse transcriptase, in which NaFM performed marginally less effectively. The slightly weaker performance of NaFM on HIV type-1 reverse transcriptase likely originated from the distinct characteristics of the molecular features. The *N*-Gram baseline, which encodes short, local SMILES substrings and substructure frequency patterns, effectively exploited recurring motif-like signals characteristic of certain antiviral chemotypes. In contrast, the ligand-based self-supervised embeddings of NaFM primarily captured higher-order scaffold topology and broader functional-group context, accounting for its state-of-the-art performance across most human protein targets. Importantly, even for HIV type-1 reverse transcriptase, our model still outperformed most baselines and remained competitive.

Moreover, NaFM substantially outperformed graph multiview pretraining (GraphMVP) and PretrainGNN across diverse protein targets, including PTP1B, AChE, COX-2, tyrosinase and CYP3A4. NaFM consistently yielded lower r.m.s. error values, such as 0.9286 on COX-1 compared with 3.0083 for GraphMVP and 2.6831 for PretrainGNN, with the advantage most pronounced for complex or data-scarce targets.

The weaker performance of GraphMVP likely stemmed from its reliance on molecular geometry during pretraining. As discussed in Supplementary Section 2.2, NPs combine rigid substructures with flexible global conformations, resulting in broad and diverse 3D distributions. This structural heterogeneity hampers geometry-based models such as GraphMVP, limiting their ability to capture the full conformational space of NPs.

We further applied the representations of NaFM to virtual screening of NPs—a critical yet challenging step in drug discovery⁵⁹. Conventional docking and deep-learning methods^{60–62} are effective for synthetic compounds but often struggle with NPs, with complex stereochemistry and reactive groups that complicate synthesis and screening⁶³.

To evaluate the robustness of NaFM across functionally and structurally diverse targets, we performed a systematic validation on AChE, a key esterase in neurodegenerative disease therapy⁶⁴. In addition, we carried out supplementary validation on protein tyrosine phosphatase 1B (PTP-1B)⁶⁵, as described in Supplementary Section 2.6.

For the AChE case study, we used the IC₅₀ data from the NPASS database, categorizing compounds as active or inactive based on a micromolar cutoff. The model was fine-tuned on the activity data and subsequently used to score the activity of compounds in the COCONUT dataset. Following the screening strategy above, we performed 100-ns molecular dynamics (MD) simulations and binding free energy (ΔG) calculations for the top 16 model-ranked compounds and three reference drugs: the AChE inhibitors (galantamine and tacrine) and memantine. Memantine, which is co-administered clinically but does not bind or inhibit AChE, was included as a negative control.

Docking poses were used as initial conformations, and 100-ns MD simulations were performed with GROMACS using the AMBER99SB-ILDN force field. Representative r.m.s. deviation profiles are shown in Supplementary Fig. 14. Binding free energies (ΔG) were estimated using molecular mechanics Poisson–Boltzmann surface area on the equilibrated trajectory segments (final 10 ns after r.m.s. deviation stabilization; Fig. 4a). Two of the 16 compounds exhibited lower ΔG than both positive controls, indicating stronger predicted binding, and the consistently low ΔG values across top-ranked molecules confirm the screening effectiveness of NaFM. Comparable PTP-1B results are reported in Supplementary Section 2.6.

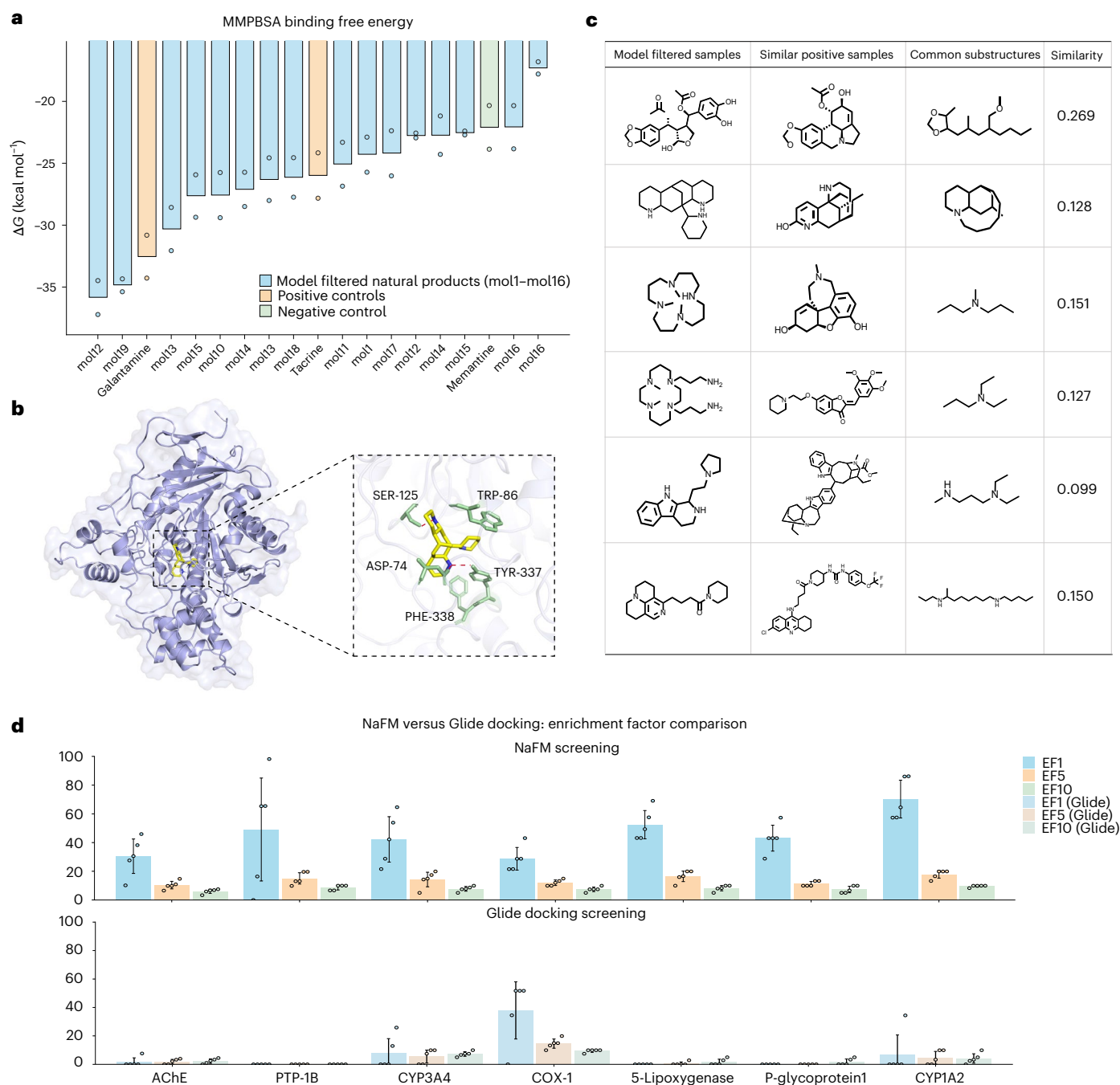


Fig. 4 | Computational validation and benchmarking of NaFM-predicted inhibitors. a, Binding free energies (ΔG) of the top 16 predicted compounds (blue), positive controls (orange: galantamine and tacrine) and negative control (light green: memantine). ΔG was estimated via MMPBSA (generalized Born model) using 100 snapshots from the final 10 ns of equilibrated trajectories. Bar heights represent the mean of $n = 2$ independent MD simulations (individual data points overlaid), performed as methodological repeats to ensure stability. **b**, Visualization of MD-simulated interactions. In the ligand, carbon atoms are shown in bright yellow and nitrogen atoms in blue. Surrounding amino acid residues are coloured green. The model-selected molecule forms three hydrogen bonds with TYR337, ASP74 and SER125 (red dashed lines) and exhibits

additional close-contact interactions with other residues, such as TRP86 and PHE338. **c**, The first and second columns display several high-scoring molecules predicted by the model, along with their most similar experimentally validated actives encountered during the fine-tuning process. The third and fourth columns illustrate the maximum common substructures between each pair of molecules and their corresponding Tanimoto similarities computed using ECFP4 fingerprints. **d**, EFs for NaFM and Glide across seven targets. The results represent mean $\pm$ s.d. from fivefold cross-validation ($n = 5$ independent splits). Bar heights and error bars denote mean and s.d., respectively. Higher values indicate superior screening performance. MMPBSA: molecular mechanics Poisson–Boltzmann surface area.

For the top-ranked compound Mol12 (lowest ΔG), an equilibrated MD snapshot was analysed in PyMOL to visualize binding (Fig. 4b). Mol12 forms a dense interaction network in the AChE active-site gorge; its ammonium group (NH_2^+) engages ASP74, SER125 and TYR337 via hydrogen bonds and is further stabilized by hydrophobic and van der

Waals contacts with TRP86 and PHE338. Beyond potency, we evaluated scaffold novelty. Structural similarity to known actives was quantified using maximum common substructure overlap and ECFP4 Tanimoto similarity (Fig. 4c). Despite a diverse fine-tuning set (44 actives and 37 Bemis–Murcko scaffolds), NaFM recovered active compounds with

low similarity to known ligands, demonstrating strong generalization and the ability to discover novel scaffolds.

To further demonstrate the capability of NaFM across diverse human protein targets, we fine-tuned the model on activity data from seven representative targets and applied the same virtual screening procedure using enrichment factors (EFs) as metrics. Glide, a widely used docking-based screening tool, was used for comparison. As shown in Fig. 4d, NaFM outperformed Glide in EF1%, EF5% and EF10% across all targets, except COX-1, in which EF1% is slightly lower. Notably, for AChE, PTP-1B, 5-lipoxygenase and P-glycoprotein 1, Glide showed minimal enrichment, whereas NaFM achieved excellent performance, likely reflecting the difficulty of fitting complex NP structures and scaffold geometries.

These results provide multitarget, physics-informed validation of the transferability of NaFM across protein families. Although broader cross-family benchmarking remains as future work, NaFM demonstrates strong potential as an efficient virtual screening framework for discovering bioactive and structurally diverse NPs.

Ablation and sensitivity analysis

We conducted a comprehensive ablation analysis to quantify the contributions of different components in NaFM, including its architectural modules, CL design and dataset composition.

Model architecture. To assess the individual effects of masked graph modelling and CL, we retrained NaFM while removing each module independently (Supplementary Table 3). The complete NaFM achieved the highest performance (AUPRC: 70.10 ± 0.92 ; accuracy: 64.78 ± 0.80). Removing masked graph modelling reduced accuracy to 57.87 ± 2.13 versus 64.78 ± 0.80 (−6.9 points), whereas excluding CL led to 61.75 ± 2.05 versus 64.78 ± 0.80 (−3.0 points). These results confirm the complementary nature of the two objectives and highlight CL as the most critical component. We also examined the influence of the encoder on model performance. The detailed results are provided in Supplementary Tables 4 and 5.

Contrastive learning design. To further investigate the contribution of the NaFM CL design, we performed ablation experiments focusing on two key components: the incorporation of scaffold-based weighting and the choice of scheduling strategies (Supplementary Table 5). Compared with a standard non-weighted contrastive set-up (AUPRC: 65.24 ± 0.86 at four samples per Class), incorporating scaffold-based weighting and dynamic scheduling improved AUPRC by nearly 5 points. Moreover, alternative scheduling strategies, such as exponential (65.64 ± 0.43) or logarithmic (67.12 ± 0.36) versus cosine (70.10 ± 0.92), consistently underperformed. These findings emphasize that the synergy between scaffold-aware weighting and cosine-based scheduling is key to the robust contrastive representation learning in NaFM.

Pretraining dataset composition. To disentangle the effects of data from methodological gains, we re-pretrained all baseline methods that support pretraining (MolCLR, PretrainGNN and Mole-BERT) on the same COCONUT dataset and evaluated them across varying data-density regimes for classification tasks (Supplementary Table 6). Re-pretrained MolCLR underperformed its original checkpoint in both low-data and high-data regimes (41.49 ± 1.05 versus 45.76 ± 1.98), whereas PretrainGNN also showed decreased performance at high-data densities (81.28 ± 0.83 versus 87.82 ± 0.13). Mole-BERT exhibited an increase in high-data performance (89.47 ± 0.67 versus 85.69 ± 0.70), but its low-data performance dropped noticeably (61.07 ± 0.58 versus 66.32 ± 1.23). Notably, regardless of whether these models were pre-trained on COCONUT, all baselines performed substantially worse than NaFM across all data regimes. These findings indicate that the observed advantages do not arise from dataset choice or data bias. Instead, they underscore that the key driver of the effectiveness of NaFM lies in the suitability of its pretraining objectives to the NP domain.

Overall, these ablations demonstrate that the superior performance of NaFM stems from the combined effects of (1) its dual pretraining objectives; (2) scaffold-aware and dynamically scheduled CL; and (3) a pretraining strategy purpose-built for NP chemistry.

Conclusion

In this paper, we proposed NaFM, a foundation model for small-molecule NPs. NaFM leverages the power of a large amount of unlabelled NP structures and an efficient biological insight-inspired pretraining strategy. Our method has been evaluated across a range of tasks, including upstream genome mining, biological source identification, structural classification, metabolic pathway prediction and downstream activity prediction and screening. On most of the benchmarks, NaFM outperformed both traditional molecular fingerprints and deep-learning-based pretraining frameworks. In the future, NaFM holds great potential as a foundation model for NP-based drug discovery and biosynthesis studies. Furthermore, it can serve as a tool-driven platform that streamlines key aspects of research workflow for NPs, integrating computational analysis, predictive modelling and experimental validation to accelerate discoveries in the field.

Methods

Initial representations of molecules

In our study, a molecule was represented as an attributed graph $G = (V, E)$, in which atoms were nodes (V) and chemical bonds were edges (E). We adopted the standard feature representation commonly used in previous studies³⁷. As detailed in Supplementary Table 1, the initial node features were constructed by summing the embeddings of attributes, including atom types, chirality and formal charges. Similarly, the initial bond features were obtained by summing the embeddings of bond types and bond directions:

$$\text{Representation}_{\text{atom/bond}} = \sum_{i=0}^{F_{\text{atom/bond}}} \text{Embed}_i(\text{Attribute}_i) \quad (1)$$

where F is the number of atom or bond features. These initial features also served as labels for the reconstruction task. We defined the possible categories as the Cartesian product of all attribute sets, resulting in $9 \times 4 \times 5 = 180$ distinct atom categories (M_{atom}) and $4 \times 3 = 12$ distinct bond categories (M_{bond}). In other words, each unique combination of atom attributes forms an atom category, and each unique pair of bond attributes defines a bond category.

Model architecture

We adopted the traditional message-passing scheme introduced in GIN⁶⁶, with modifications to incorporate edge features, as proposed in a previous study⁶⁷. The node representation is updated as follows:

$$\hat{\mathbf{h}}_i^{l+1} = \text{MLP} \left(\mathbf{h}_i^l + \sum_{j \in \mathcal{N}(i)} (\mathbf{h}_j^l + \mathbf{e}_{ij}) \right), \quad (2)$$

where $\mathbf{h}_i^l$ is the representation of node i in the l th layer, $\mathbf{e}_{ij}$ represents the edge feature between nodes i and j and $\mathcal{N}(i)$ denotes the neighbours of node i . To stabilize training, we applied layer normalization after each GIN block:

$$\mathbf{h}_i^{l+1} = \text{LayerNorm}(\hat{\mathbf{h}}_i^{l+1}) \quad (3)$$

The final node representations, $\mathbf{h}^L$, are aggregated using an average pooling operation to derive the graph-level representation:

$$\mathbf{h}_{\text{graph}} = \frac{1}{|V(G)|} \sum_{v \in G} \mathbf{h}_v^L \quad (4)$$

where $|V(G)|$ is the number of nodes in each graph. For the CL task, an additional multilayer perceptron (MLP) was used to project the graph

representation into the latent space. For reconstruction, the final node representations were mapped to the corresponding atom categories, whereas the final edge representations were obtained by concatenating the node representations and applying an MLP to adapt them to the specific tasks. During fine-tuning on downstream tasks, we removed the CL and reconstruction head and attached an MLP to the final graph representation for classification or regression tasks.

Scaffold-aware contrastive learning. The issue of mislabelling in CL can be mitigated through the application of reweighting techniques⁶⁸. However, determining appropriate weights is a challenging task. Interestingly, the distinctive characteristics of NPs—specifically, the strong correlation between their scaffold structures and their properties—offer a promising solution for weight selection: scaffold similarity. To implement this, we computed the cosine similarity between the Molecular Access System fingerprints⁶⁹ of their scaffolds. Specifically, scaffolds were defined here according to the Bemis–Murcko framework. This similarity was subsequently incorporated into the CL loss function for reweighting, as detailed below:

$$\mathcal{L}_c^w = \frac{1}{N} \sum_{m=1}^N [\ell(2m-1, 2m) + \ell(2m, 2m-1)], \ell_{ij} = -\log \frac{s_{ij}}{s_{ij} + \sum_{k=1}^{2N} \mathbb{I}_{[k \neq i, j]} (1 - w_{i,k}) s_{i,k}} \quad (5)$$

where $s_{ij} = \exp(\text{sim}(h_i, h_j)/\tau)$ is the similarity in exponential space between the representations of the i th and j th molecules in a batch containing N samples, and $\mathbb{I}$ is the indicator function.

The weighted loss function can be interpreted as assigning lower weights to negative sample pairs from different molecules that exhibit more scaffold similarity. This allows the model to reduce its attention to these ‘false negatives’, preventing it from being misled. As a result, the model achieves a more robust representation of molecular topology structures. Although reweighting techniques were introduced, the model tended to overfit the CL task as training progressed, leading to attempts to distinguish these false negatives. To further address this issue, we developed a cosine loss weight scheduler as follows:

$$\lambda_c^i = \frac{1}{2} \cos\left(1 + \frac{i\pi}{N}\right) \quad (6)$$

where i is the current epoch, and N is the total epochs. In this way, the model can mitigate overfitting on this task while ensuring sufficient training for other tasks.

Scaffold-subgraph reconstruction. With the data-augmented samples, which naturally provide labels for mask modelling, the two frameworks were seamlessly integrated. These two approaches are complementary; CL is prone to overfitting, as contrastive models may find shortcuts through trivial representations. However, with the aid of mask modelling, the model goes beyond merely distinguishing between positive and negative sample pairs. It leverages information from functional groups (non-masked) to predict scaffold information (local). Simultaneously, CL enhances the model by helping it learn a global semantic representation. It is important to note that we defined the task as scaffold-subgraph reconstruction while discarding traditional approaches, such as atomic masking and bond deletion. These tasks are considered too simplistic for the model and do not contribute meaningful knowledge. To ensure that the model focuses on scaffold modelling, we limited the masked portions exclusively to the atoms within the scaffold. In the reconstruction phase, we introduced a loss function that combines three classification tasks: atom type prediction, link prediction and bond type prediction:

$$\mathcal{L}_r = \mathcal{L}_{\text{atom}} + \mathcal{L}_{\text{link}} + \mathcal{L}_{\text{bond}} \quad (7)$$

$$= -\frac{1}{N_{\text{atom}}} \sum_i \sum_{c=1}^{M_{\text{atom}}} y_{ic} \log(p_{\text{atom}}^{ic}) - \frac{1}{N_{\text{bond}}} \sum_j \sum_{c=1}^2 y_{jc} \log(p_{\text{link}}^{jc}) - \frac{1}{N_{\text{bond}}} \sum_j \sum_{c=1}^{M_{\text{bond}}} y_{jc} \log(p_{\text{bond}}^{jc}) \quad (8)$$

where N_* is the number of atoms or bonds, and M_* is the number of atom or bond category. Here y_{ic} represents the indicator function, where the value is 1 if sample i belongs to category c , and 0 otherwise. Additionally, p_*^{ic} denotes the probability that the model predicts sample i that belongs to category c . More detailed information about atom or bond categories is presented in Methods.

Scaffold masking algorithm

Scaffold masking is a critical step in our framework, as it plays a key role in both the augmented data of CL and the reconstruction task. The masking process is described in detail below.

As shown in Algorithm 1, initially, one scaffold atom is randomly selected, and its corresponding node is removed. The process continues by identifying the neighbours of the currently selected nodes, prioritizing those in the scaffold. These neighbouring nodes are shuffled and iteratively removed until the desired number of nodes is reached or no further eligible neighbours exist. The output is a tensor containing the indices of the removed nodes. After that, all removed nodes are marked as masked, and all edges containing any of the removed nodes are also masked to form an augmented graph of CL and input–output pair of the reconstruction task.

Algorithm 1. Masked subgraph generation

Algorithm 1: Masked Subgraph Generation

Input: Graph G , Scaffold IDs S , Mask ratio m

Output: Modified graph G_{new} , Masked nodes V_{masked}

1 Initialization:

$M \leftarrow \max(\min(|V(G)| \cdot m, |S|), 2)$

$V_{\text{masked}} \leftarrow \emptyset$

$V_{\text{temp}} \leftarrow \text{Random choice}(S, 1)$

2 while $|V_{\text{masked}}| < M$ do

3 $V_{\text{neighbors}} \leftarrow \emptyset$

4 **foreach** $n \in V_{\text{temp}}$ **do**

5 $V_{\text{neighbors}} \leftarrow V_{\text{neighbors}} \cup \{i \in \text{Neighbors}(G, n) \mid i \notin V_{\text{temp}} \text{ and } i \in S\}$

6 **end foreach**

7 $V_{\text{neighbors}} \leftarrow \text{Shuffle}(V_{\text{neighbors}})$

8 **foreach** $n \in V_{\text{temp}}$ **do**

9 **if** $|V_{\text{masked}}| < M$ **then**

10 $G \leftarrow \text{Remove node}(G, n)$

11 $V_{\text{masked}} \leftarrow V_{\text{masked}} \cup \{n\}$

12 **end if**

13 **end foreach**

14 $V_{\text{temp}} \leftarrow \text{unique nodes in } V_{\text{neighbors}}$

15 **if** $V_{\text{temp}} = \emptyset$ **then**

16 **break**

17 **end if**

18 **end while**

19 **return** V_{masked}

Training details

Pretraining. We implemented a six-layer GIN¹⁶⁶ with layer normalization as the GNN backbone, using a hidden dimensionality of 512. An average pooling operation was applied as the read-out function to extract a global molecular representation of 1,024 dimensions. For the CL task, a single MLP with LeakyReLU activations projected the representation to 256 dimensions. For the reconstruction task, three MLPs with LeakyReLU activations mapped the node or edge representations to their respective output dimensions. All MLPs used a hidden dimensionality of 512. The masking ratio was set to 0.2.

We optimized the weighted CL and reconstruction loss using the AdamW optimizer with a weight decay of 10^{-5} . The temperature for the CL loss τ was set to 0.1. A cosine decay schedule was applied to the scaling factor of the CL loss, decreasing from 1 to 0. The learning rate was set to 1×10^{-4} with a cosine decay schedule. The model was trained with a batch size of 256 for a total of 300 epochs. To avoid overfitting, we also set the dropout ratio to 0.1. A detailed hyperparameter search space is provided in Supplementary Table 2. To pretrain NaFM, we employed four NVIDIA A100 graphics processing units and trained the model continuously for 5 days.

Downstream fine-tuning. For downstream task fine-tuning, we replaced the original CL head and reconstruction head with an additional, randomly initialized MLP on top of the GNN backbone. The softmax cross-entropy loss was used for classification tasks. For each task, we used the AdamW optimizer with a weight decay of 1×10^{-5} and a learning rate of 5×10^{-4} , along with a cosine learning rate decay schedule. The model was fine-tuned for 300 epochs with a batch size of 512. The dropout ratio in the added MLP was set to 0.3. Early stopping was applied if the evaluation metric did not improve for 20 consecutive epochs. Thanks to the robust performance of pretraining, we did not perform hyperparameter tuning for the downstream tasks. For the data splitting strategy, we employed stratified splits for the taxonomy classification of NPs to ensure balanced class distribution across the training, validation and test sets. For the biological source, Pfam prediction and bioactivity prediction tasks, we used random splits. For all pretrained baselines, we used the same settings as described above. For baselines trained from scratch, we followed their default configurations.

Data availability

The minimum datasets required to interpret, verify and extend the findings of this study are publicly available via figshare at <https://doi.org/10.6084/m9.figshare.28980254.v1> (ref. 70). The repository contains the following datasets used for pretraining and downstream evaluation: (1) pretrain_smiles.pkl: preprocessed data used for model pretraining, with the original data obtained from the COCONUT database (<https://coconut.naturalproducts.net/>); (2) classification_data.csv: data prepared for the NP taxonomy classification experiment, with the original dataset sourced from <https://doi.org/10.26434/chemrxiv.12885494.v1> (ref. 71); (3) NPClassifier_dataset_refreshed.csv: data curated for direct comparison with the NP Classifier, with the original data available at https://github.com/mwang87/NP-Classifier/tree/master/training/Data/NPClassifier_dataset.xlsx; (4) regression_data.csv: dataset used for NP bioactivity prediction tasks, with the original data retrieved from the NPASS database (<https://bidd.group/NPASS/>); (5) lotus_data.csv: data prepared for biological source prediction and mining, with the source data collected from the LOTUS database (<https://lotus.naturalproducts.net/>); (6) bgc_data.csv: dataset constructed for BGC mining, with the original sources included in the MIBiG database (<https://mibig.secondarymetabolites.org/>) and Pfam (<http://pfam.xfam.org/>); and (7) external_data.csv: dataset used for bioactivity screening of NPs, with the original data obtained from the NPASS database (<https://bidd.group/NPASS/>) and the pretrained weights are available via Zenodo at <https://doi.org/10.5281/zenodo.15385335> (ref. 72). Source data are provided with this paper.

Code availability

The source code used in this study is publicly available at <https://github.com/TomAIDD/NaFM-Official>. An archived version corresponding to the code used in this paper is available via Zenodo at <https://doi.org/10.5281/zenodo.18871560> (ref. 73).

References

- Chandrasekhar, V. et al. COCONUT 2.0: a comprehensive overhaul and curation of the collection of open natural products database. *Nucleic Acids Res.* **53**, 634–643 (2025).
- Newman, D. J. & Cragg, G. M. Natural products as sources of new drugs from 1981 to 2014. *J. Nat. Prod.* **79**, 629–661 (2016).
- Clark, A. M. Natural products as a resource for new drugs. *Pharm. Res.* **13**, 1133–1141 (1996).
- Harvey, A. L. Natural products in drug discovery. *Drug Discov. Today* **13**, 894–901 (2008).
- Li, J. W.-H. & Vederas, J. C. Drug discovery and natural products: end of an era or an endless frontier? *Science* **325**, 161–165 (2009).
- Atanasov, A. G., Zotchev, S. B., Dirsch, V. M. & Supuran, C. T. Natural products in drug discovery: advances and opportunities. *Nat. Rev. Drug Discov.* **20**, 200–216 (2021).
- Corson, T. W. & Crews, C. M. Molecular understanding and modern application of traditional medicines: triumphs and trials. *Cell* **130**, 769–774 (2007).
- Irwin, J. J. et al. ZINC20—a free ultralarge-scale chemical database for ligand discovery. *J. Chem. Inf. Model.* **60**, 6065–6073 (2020).
- Enamine REAL Database: The Largest Enumerated Dataset of Synthetically Feasible Drug-like Molecules (Enamine, accessed 7 October 2025); <https://enamine.net/compound-collections/real-compounds/real-database>
- Stokes, J. M. et al. A deep learning approach to antibiotic discovery. *Cell* **180**, 688–702 (2020).
- Banerjee, P. et al. Super Natural II—a database of natural products. *Nucleic Acids Res.* **43**, 935–939 (2015).
- Sorokina, M., Merseburger, P., Rajan, K., Yirik, M. A. & Steinbeck, C. COCONUT online: collection of open natural products database. *J. Cheminform.* <https://doi.org/10.1186/s13321-020-00478-9> (2021).
- Rutz, A. et al. The LOTUS initiative for open knowledge management in natural products research. *Elife* **11**, e70780 (2022).
- Zeng, X. et al. NPASS: natural product activity and species source database for natural product research, discovery and tool development. *Nucleic Acids Res.* **46**, 1217–1222 (2018).
- van Santen, J. A. et al. The Natural Products Atlas: an open access knowledge base for microbial natural products discovery. *ACS Cent. Sci.* **5**, 1824–1833 (2019).
- Terlouw, B. R. et al. MIBiG 3.0: a community-driven effort to annotate experimentally validated biosynthetic gene clusters. *Nucleic Acids Res.* **51**, 603–610 (2023).
- Lei, J. & Zhou, J. A marine natural product database. *J. Chem. Inf. Comput. Sci.* **42**, 742–748 (2002).
- Barbosa, A. J. & Roque, A. C. Free marine natural products databases for biotechnology and bioengineering. *Biotechnol. J.* **14**, 1800607 (2019).
- Lyu, C. et al. CMNPD: a comprehensive marine natural products database towards facilitating drug discovery from the ocean. *Nucleic Acids Res.* **49**, 509–515 (2021).
- Aghdam, S. A. & Brown, A. M. V. Deep learning approaches for natural product discovery from plant endophytic microbiomes. *Environ. Microbiome* **16**, 6 (2021).
- Zheng, S. et al. Deep learning driven biosynthetic pathways navigation for natural products with BioNavi-NP. *Nat. Commun.* <https://doi.org/10.1038/s41467-022-30970-9> (2022).
- Lai, J. et al. Privileged scaffold analysis of natural products with deep learning-based indication prediction model. *Mol. Inform.* **39**, e2000057 (2020).
- Yoo, S. et al. A deep learning-based approach for identifying the medicinal uses of plant-derived natural compounds. *Front. Pharmacol.* **11**, 584875 (2020).
- Hannigan, G. D. et al. A deep learning genome-mining strategy for biosynthetic gene cluster prediction. *Nucleic Acids Res.* **47**, e110 (2019).
- Liu, Z. et al. Deep learning enables discovery of highly potent anti-osteoporosis natural products. *Eur. J. Med. Chem.* **210**, 112982 (2021).

26. Xu, Q. et al. Composite machine learning strategy for natural products taxonomical classification and structural insights. *Digital Discov.* **3**, 2192–2200 (2024).
27. Djoumbou Feunang, Y. et al. ClassyFire: automated chemical classification with a comprehensive, computable taxonomy. *J. Cheminform.* **8**, 1–20 (2016).
28. Kim, H. W. et al. NPClassifier: a deep neural network-based structural classification tool for natural products. *J. Nat. Prod.* **84**, 2795–2807 (2021).
29. Yu, L., Su, Y., Liu, Y. & Zeng, X. Review of unsupervised pretraining strategies for molecules representation. *Brief. Funct. Genomics* **20**, 323–332 (2021).
30. Weininger, D., Weininger, A. & Weininger, J. L. SMILES. 2. Algorithm for generation of unique SMILES notation. *J. Chem. Inf. Comput. Sci.* **29**, 97–101 (1989).
31. Cho, K. et al. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proc. 2014 Conference on Empirical Methods in Natural Language Processing* 1724–1734 (2014).
32. Vaswani, A. et al. Attention is all you need. In *Proc. 30th International Conference on Advances in Neural Information Processing Systems* (eds Guyon, I. et al.) 6000–6010 (Curran, 2017).
33. Xu, Z., Wang, S., Zhu, F. & Huang, J. Seq2seq fingerprint: an unsupervised deep molecular embedding for drug discovery. In *Proc. 8th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics* (eds Haspel, N. et al.) 285–294 (Association for Computing Machinery, 2017).
34. Jastrzębski, S., Leśniak, D. & Czarnecki, W. M. Learning to SMILE(S). Preprint at <https://arxiv.org/abs/1602.06289> (2016).
35. Kearnes, S., McCloskey, K., Berndl, M., Pande, V. & Riley, P. Molecular graph convolutions: moving beyond fingerprints. *J. Comput.-Aided Mol. Des.* **30**, 595–608 (2016).
36. Schütt, K. et al. SchNet: a continuous-filter convolutional neural network for modeling quantum interactions. In *Proc. 31st Conference on Neural Information Processing Systems* (eds von Luxburg, U. et al.) 992–1002 (Curran Associates, 2017).
37. Wang, Y., Wang, J., Cao, Z. & Barati Farimani, A. Molecular contrastive learning of representations via graph neural networks. *Nat. Mach. Intell.* **4**, 279–287 (2022).
38. Hu, W. et al. Strategies for pre-training graph neural networks. In *International Conference on Learning Representations* (OpenReview.net, 2020).
39. Xia, J. et al. Mole-BERT: rethinking pre-training graph neural networks for molecules. In *International Conference on Learning Representations* <https://openreview.net/pdf?id=jevY-DtiZTR> (2023).
40. Liu, S. et al. Pre-training molecular graph representation with 3D geometry. Preprint at <https://arxiv.org/abs/2110.07728> (2021).
41. Zhu, J. et al. Unified 2D and 3D pre-training of molecular representations. In *Proc. 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (eds Zhang, A.) 2626–2636 (Association for Computing Machinery, 2022).
42. Li, H. et al. A knowledge-guided pre-training framework for improving molecular representation learning. *Nat. Commun.* <https://doi.org/10.1038/s41467-023-43214-1> (2023).
43. Ni, Y. et al. Pre-training with fractional denoising to enhance molecular property prediction. *Nat. Mach. Intell.* **6**, 1169–1178 (2024).
44. Mullowney, M. W. et al. Artificial intelligence for natural product drug discovery. *Nat. Rev. Drug Discov.* **22**, 895–916 (2023).
45. Garcia-Castro, M., Zimmermann, S., Sankar, M. G. & Kumar, K. Scaffold diversity synthesis and its application in probe and drug discovery. *Angew. Chem. Int. Ed.* **55**, 7586–7605 (2016).
46. Cruz-Monteagudo, M. et al. Activity cliffs in drug discovery: Dr Jekyll or Mr Hyde? *Drug Discov. Today* **19**, 1069–1080 (2014).
47. Stumpfe, D., Hu, H. & Bajorath, J. Evolving concept of activity cliffs. *ACS Omega* **4**, 14360–14368 (2019).
48. van Tilborg, D., Alenicheva, A. & Grisoni, F. Exposing the limitations of molecular machine learning with activity cliffs. *J. Chem. Inf. Model.* **62**, 5938–5951 (2022).
49. Shen, W. X. et al. Online triplet contrastive learning enables efficient cliff awareness in molecular activity prediction. Preprint at *ChemRxiv* <https://doi.org/10.26434/chemrxiv-2023-5cz7s-v2> (2023).
50. Sun, R., Dai, H. & Yu, A. W. Does GNN pretraining help molecular representation? In *Proc. 36th International Conference on Neural Information Processing Systems* (eds Koyejo, S. et al.) 12096–12109 (Curran, 2022).
51. Martin, J. F. & Liras, P. Organization and expression of genes involved in the biosynthesis of antibiotics and other secondary metabolites. *Annu. Rev. Microbiol.* **43**, 173–206 (1989).
52. Martin, J. F. Clusters of genes for the biosynthesis of antibiotics: regulatory genes and overproduction of pharmaceuticals. *J. Ind. Microbiol.* **9**, 73–90 (1992).
53. Carroll, L. M. et al. Accurate de novo identification of biosynthetic gene clusters with GECCO. Preprint at *bioRxiv* <https://doi.org/10.1101/2021.05.03.442509> (2021).
54. Sanchez, S. et al. Expansion of novel biosynthetic gene clusters from diverse environments using SanntiS. Preprint at *bioRxiv* <https://doi.org/10.1101/2023.05.23.540769> (2023).
55. Mistry, J. et al. Pfam: the protein families database in 2021. *Nucleic Acids Res.* **49**, 412–419 (2021).
56. Marchler-Bauer, A. et al. CDD: a conserved domain database for interactive domain family analysis. *Nucleic Acids Res.* **35**, 237–240 (2007).
57. Ulrich, L. E. & Zhulin, I. B. The MiST2 database: a comprehensive genomics resource on microbial signal transduction. *Nucleic Acids Res.* **38**, 401–407 (2010).
58. Zeng, T., Li, J. & Wu, R. Natural product databases for drug discovery: features AND applications. *Pharm. Sci. Adv.* **2**, 100050 (2024).
59. Maia, E. H. B., Assis, L. C., de Oliveira, T. A., da Silva, A. M. & Taranto, A. G. Structure-based virtual screening: from classical to artificial intelligence. *Front. Chem.* **8**, 343 (2020).
60. Friesner, R. A. et al. Glide: a new approach for rapid, accurate docking and scoring. 1. Method and assessment of docking accuracy. *J. Med. Chem.* **47**, 1739–1749 (2004).
61. Trott, O. & Olson, A. J. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J. Comput. Chem.* **31**, 455–461 (2010).
62. Kimber, T. B., Chen, Y. & Volkamer, A. Deep learning in virtual screening: recent applications and developments. *Int. J. Mol. Sci.* **22**, 4435 (2021).
63. Ma, D.-L., Chan, D. S.-H. & Leung, C.-H. Molecular docking for virtual screening of natural product databases. *Chem. Sci.* **2**, 1656–1665 (2011).
64. Soreq, H. & Seidman, S. Acetylcholinesterase—new roles for an old actor. *Nat. Rev. Neurosci.* **2**, 294–302 (2001).
65. Delibegović, M., Dall’Angelo, S. & Dekeryte, R. Protein tyrosine phosphatase 1B in metabolic diseases and drug development. *Nat. Rev. Endocrinol.* **20**, 366–378 (2024).
66. Xu, K., Hu, W., Leskovec, J. & Jegelka, S. How powerful are graph neural networks? In *International Conference on Learning Representations* <https://openreview.net/pdf?id=ryGs6iA5Km> (2019).
67. Hu, W. et al. OGB-LSC: a large-scale challenge for machine learning on graphs. In *Proc. of the 35th Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)* (eds Vanschoren, J. & Yeung, S.) <https://openreview.net/pdf?id=qkcLxoC52kL> (2021).

68. Wang, Y., Magar, R., Liang, C. & Barati Farimani, A. Improving molecular contrastive learning via faulty negative mitigation and decomposed fragment contrast. *J. Chem. Inf. Model.* **62**, 2713–2725 (2022).
69. Durant, J. L., Leland, B. A., Henry, D. R. & Nourse, J. G. Reoptimization of MDL keys for use in drug discovery. *J. Chem. Inf. Comput. Sci.* **42**, 1273–1280 (2002).
70. Ding, Y. et al. Data archiving and access for NaFM: pre-training a foundation model for small-molecule natural products. *figshare* <https://doi.org/10.6084/m9.figshare.28980254.v1> (2025).
71. Kim, H. et al. NPClassifier: a deep neural network-based structural classification tool for natural products. *J. Nat. Prod.* **84**, 2795–2807 (2021).
72. Ding, Y. et al. Model weights for NaFM: pre-training a foundation model for small-molecule natural products. *Zenodo* <https://doi.org/10.5281/zenodo.15382660> (2025).
73. Ding, Y. et al. NaFM-Official: version 1.0.0. *Zenodo* <https://doi.org/10.5281/zenodo.18871560> (2025).
74. Liu, S., Demirel, M. F. & Liang, Y. N-gram graph: simple unsupervised representation for graphs, with applications to molecules. In *Proc. 33rd Conference on Neural Information Processing Systems* (eds Wallach, H. et al.) 8464–8476 (Curran Associates, 2019).
75. Yang, K. et al. Analyzing learned molecular representations for property prediction. *J. Chem. Inf. Model.* **59**, 3370–3388 (2019).
76. Rogers, D. & Hahn, M. Extended-connectivity fingerprints. *J. Chem. Inf. Model.* **50**, 742–754 (2010).

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Author contributions

Y.D. and B.Q. conceived the project. Y.D. and Y.W. processed the dataset and trained the model. Both performed computational downstream tasks, and Y.D. also conducted biochemical analyses. Y.Z. and J.Y. provided support for specific downstream datasets. Y.D. and B.Q. analysed the results. Y.D. wrote the original draft of the paper. Y.D., B.Q., S.L., J.Y., C.S., Q.L. and L.Z. contributed to the revision of the paper. The project was supervised by Z.L., N.Z., Y.W. and B.Q. All authors participated in discussions and provided critical feedback.

Competing interests

The authors declare no competing interests.

Additional information

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